

Daraz E-Commerce Database

Problem Statement:

A simplified relational database for **Daraz Bangladesh** is provided, consisting of four tables: Customers, Products, Orders, and OrderDetails. Based on the given table structures, answer the following SQL-related queries.

Table Structures:

Task 1: Database and Table Creation

1. **Create a new database** named **DarazBD**.
2. **Create a table** named **Customers** with the following fields and constraints:

Field	Data Type	Constraints
customer_id	INT	PRIMARY KEY, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	UNIQUE, NOT NULL
phone	VARCHAR(15)	NOT NULL
address	VARCHAR(255)	NULL
registration_date	DATE	NOT NULL

3. **Create a table** named **Products** with the following fields and constraints:

Field	Data Type	Constraints
product_id	INT	PRIMARY KEY, AUTO_INCREMENT

product_name	VARCHAR(100)	NOT NULL
category	VARCHAR(50)	NOT NULL
price	DECIMAL(10,2)	NOT NULL
stock_qty	INT	NOT NULL

4. **Create a table** named **Orders** with the following fields and constraints:

Field	Data Type	Constraints
order_id	INT	PRIMARY KEY, AUTO_INCREMENT
customer_id	INT	FOREIGN KEY REFERENCES Customers(customer_id)
order_date	DATE	NOT NULL
total_amount	DECIMAL(10,2)	NOT NULL
status	ENUM('Pending', 'Shipped', 'Delivered', 'Cancelled')	DEFAULT 'Pending'

5. **Create a table** named **OrderDetails** with the following fields and constraints:

Field	Data Type	Constraints
order_detail_id	INT	PRIMARY KEY, AUTO_INCREMENT
order_id	INT	FOREIGN KEY REFERENCES Orders(order_id)
product_id	INT	FOREIGN KEY REFERENCES Products(product_id)
quantity	INT	NOT NULL
unit_price	DECIMAL(10,2)	NOT NULL

Task 2: Data Insertion

Insert sample records into the respective tables using SQL INSERT INTO statements based on the data provided below.

Customers Table:

name	email	phone	address	registration_date
Ayesha Akter	ayesha@example.com	01711001122	Dhaka, Bangladesh	2024-01-05
Rahim Uddin	rahim@example.com	01888002233	Chattogram, Bangladesh	2024-02-10
Tania Sultana	tania@example.com	01999003344	Rajshahi, Bangladesh	2023-11-20
Nasir Hossain	nasir@example.com	01777004455	Khulna, Bangladesh	2024-04-01

6. Write the SQL insertion query to insert these records into the **Customers** table.

Products Table:

product_name	category	price	stock_qty
Samsung Galaxy A14	Smartphones	18999.00	20
Logitech Wireless Mouse	Accessories	950.00	50
HP Pavilion Laptop	Computers	69990.00	10
Redmi Note 12	Smartphones	17999.00	15
Sony Headphones	Accessories	2499.00	35

7. Write the SQL insertion query to insert these records into the **Products** table.

Orders Table:

customer_id	order_date	total_amount	status
1	2024-04-15	18999.00	Delivered
2	2024-04-20	27949.00	Pending
3	2024-04-25	950.00	Shipped
1	2024-05-01	2499.00	Delivered

8. Write the SQL insertion query to insert these records into the **Orders** table.

OrderDetails Table:

order_id	product_id	quantity	unit_price
1	1	1	18999.00
2	4	1	17999.00
2	5	1	9949.00
3	2	1	950.00
4	5	1	2499.00

9. Write the SQL insertion query to insert these records into the **OrderDetails** table.

QUERIES:

- A) Display all customers.
- B) List all products with price greater than 5000.
- C) Find all orders placed in April 2024.
- D) Count the number of products in each category.

- E) List delivered orders with customer names.
- F) Find customers who registered after January 2024.
- G) Total quantity sold per product.
- H) Find all customers who bought 'Sony Headphones'.
- I) List all pending orders with product details.
- J) Find the product with the highest total revenue.

SOLUTION:

```
2  -- Task 1: Database and Table Creation
3  • CREATE DATABASE DarazBD;
4  • USE DarazBD;
5
6  • CREATE TABLE Customers (
7      customer_id INT PRIMARY KEY AUTO_INCREMENT,
8      name VARCHAR(100) NOT NULL,
9      email VARCHAR(100) UNIQUE NOT NULL,
10     phone VARCHAR(15) NOT NULL,
11     address VARCHAR(255),
12     registration_date DATE NOT NULL
13 );

15 • CREATE TABLE Products (
16     product_id INT PRIMARY KEY AUTO_INCREMENT,
17     product_name VARCHAR(100) NOT NULL,
18     category VARCHAR(50) NOT NULL,
19     price DECIMAL(10,2) NOT NULL,
20     stock_qty INT NOT NULL
21 );

23 • CREATE TABLE Orders (
24     order_id INT PRIMARY KEY AUTO_INCREMENT,
25     customer_id INT,
26     order_date DATE NOT NULL,
27     total_amount DECIMAL(10,2) NOT NULL,
28     status ENUM('Pending', 'Shipped', 'Delivered', 'Cancelled') DEFAULT 'Pending',
29     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
30 );
```

```

32 ● CREATE TABLE OrderDetails (
33     order_detail_id INT PRIMARY KEY AUTO_INCREMENT,
34     order_id INT,
35     product_id INT,
36     quantity INT NOT NULL,
37     unit_price DECIMAL(10,2) NOT NULL,
38     FOREIGN KEY (order_id) REFERENCES Orders(order_id),
39     FOREIGN KEY (product_id) REFERENCES Products(product_id)
40 );
41
42 -- Task 2: Data Insertion
43 ● INSERT INTO Customers (name, email, phone, address, registration_date) VALUES
44 ('Ayesha Akter', 'ayesha@example.com', '01711001122', 'Dhaka, Bangladesh', '2024-01-05'),
45 ('Rahim Uddin', 'rahim@example.com', '01888002233', 'Chattogram, Bangladesh', '2024-02-10'),
46 ('Tania Sultana', 'tania@example.com', '01999003344', 'Rajshahi, Bangladesh', '2023-11-20'),
47 ('Nasir Hossain', 'nasir@example.com', '01777004455', 'Khulna, Bangladesh', '2024-04-01');
48
49 ● INSERT INTO Products (product_name, category, price, stock_qty) VALUES
50 ('Samsung Galaxy A14', 'Smartphones', 18999.00, 20),
51 ('Logitech Wireless Mouse', 'Accessories', 950.00, 50),
52 ('HP Pavilion Laptop', 'Computers', 69990.00, 10),
53 ('Redmi Note 12', 'Smartphones', 17999.00, 15),
54 ('Sony Headphones', 'Accessories', 2499.00, 35);
55
56 ● INSERT INTO Orders (customer_id, order_date, total_amount, status) VALUES
57 (1, '2024-04-15', 18999.00, 'Delivered'),
58 (2, '2024-04-20', 27949.00, 'Pending'),
59 (3, '2024-04-25', 950.00, 'Shipped'),
60 (1, '2024-05-01', 2499.00, 'Delivered');
61
62 ● INSERT INTO OrderDetails (order_id, product_id, quantity, unit_price) VALUES
63 (1, 1, 1, 18999.00),
64 (2, 4, 1, 17999.00),
65 (2, 5, 1, 9949.00),
66 (3, 2, 1, 950.00),
67 (4, 5, 1, 2499.00);

```

```

69      -- Task 3: SQL Queries (A-J)
70      -- A) Display all customers
71      •   SELECT * FROM Customers;

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	customer_id	name	email	phone	address	registration_date
▶	1	Ayesha Akter	ayesha@example.com	01711001122	Dhaka, Bangladesh	2024-01-05
	2	Rahim Uddin	rahim@example.com	01888002233	Chattogram, Bangladesh	2024-02-10
	3	Tania Sultana	tania@example.com	01999003344	Rajshahi, Bangladesh	2023-11-20
	4	Nasir Hossain	nasir@example.com	01777004455	Khulna, Bangladesh	2024-04-01
•	NULL	NULL	NULL	NULL	NULL	NULL

```

73      -- B) List all products with price greater than 5000
74      •   SELECT * FROM Products WHERE price > 5000;

```

Result Grid

Filter Rows:

Edit:

Export/Import:

	product_id	product_name	category	price	stock_qty
▶	1	Samsung Galaxy A14	Smartphones	18999.00	20
	3	HP Pavilion Laptop	Computers	69990.00	10
	4	Redmi Note 12	Smartphones	17999.00	15
*	NULL	NULL	NULL	NULL	NULL

```

76      -- C) Find all orders placed in April 2024
77      •   SELECT * FROM Orders WHERE order_date BETWEEN '2024-04-01' AND '2024-04-30';

```

Result Grid

Filter Rows:

Edit:

Export/Import:

	order_id	customer_id	order_date	total_amount	status
▶	1	1	2024-04-15	18999.00	Delivered
	2	2	2024-04-20	27949.00	Pending
	3	3	2024-04-25	950.00	Shipped
✱	NULL	NULL	NULL	NULL	NULL

```

-- D) Count number of products in each category
SELECT category, COUNT(*) AS total_products FROM Products GROUP BY category;

```

Result Grid	Filter Rows:
category	total_products
Accessories	2
Computers	1
Smartphones	2

```

82      -- E) List delivered orders with customer names
83      • SELECT o.order_id, c.name, o.order_date, o.status
84      FROM Orders o
85      JOIN Customers c ON o.customer_id = c.customer_id
86      WHERE o.status = 'Delivered';

```

Result Grid

Filter Rows:

Export:

	order_id	name	order_date	status
▶	1	Ayesha Akter	2024-04-15	Delivered
	4	Ayesha Akter	2024-05-01	Delivered

```

88      -- F) Find customers who registered after January 2024
89      • SELECT * FROM Customers WHERE registration_date > '2024-01-31';

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	customer_id	name	email	phone	address	registration_date
▶	2	Rahim Uddin	rahim@example.com	01888002233	Chattogram, Bangladesh	2024-02-10
	4	Nasir Hossain	nasir@example.com	01777004455	Khulna, Bangladesh	2024-04-01
*	NULL	NULL	NULL	NULL	NULL	NULL

```

91      -- G) Total quantity sold per product
92      • SELECT p.product_name, SUM(od.quantity) AS total_quantity
93      FROM OrderDetails od
94      JOIN Products p ON od.product_id = p.product_id
95      GROUP BY p.product_name;

```

Result Grid

Filter Rows:

Export:

	product_name	total_quantity
▶	Logitech Wireless Mouse	1
	Redmi Note 12	1
	Samsung Galaxy A14	1
	Sony Headphones	2

```

97      -- H) Find all customers who bought 'Sony Headphones'
98  •    SELECT DISTINCT c.name
99      FROM Customers c
100     JOIN Orders o ON c.customer_id = o.customer_id
101     JOIN OrderDetails od ON o.order_id = od.order_id
102     JOIN Products p ON od.product_id = p.product_id
103     WHERE p.product_name = 'Sony Headphones';

```

Result Grid		Filter Rows:
	name	
▶	Ayesha Akter	
	Rahim Uddin	

```

105     -- I) List all pending orders with product details
106  •    SELECT o.order_id, p.product_name, od.quantity, o.status
107      FROM Orders o
108     JOIN OrderDetails od ON o.order_id = od.order_id
109     JOIN Products p ON od.product_id = p.product_id
110     WHERE o.status = 'Pending';

```

Result Grid




Filter Rows:

Ex

	order_id	product_name	quantity	status
▶	2	Redmi Note 12	1	Pending
	2	Sony Headphones	1	Pending

```

112     -- J) Find product with the highest total revenue
113  •    SELECT p.product_name, SUM(od.quantity * od.unit_price) AS total_revenue
114      FROM OrderDetails od
115     JOIN Products p ON od.product_id = p.product_id
116     GROUP BY p.product_name
117     ORDER BY total_revenue DESC
118     LIMIT 1;

```

Result Grid		Filter Rows:
	product_name	total_revenue
▶	Samsung Galaxy A14	18999.00